

Quantum Algorithms for Cybersecurity, Chemistry, and Optimization

KEY IMAGES: Week 1

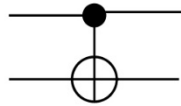
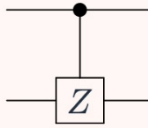
Introduction > Classical and Quantum Gates from Course 1

GATE		CIRCUIT REPRESENTATION	TRUTH TABLE										
NOT	The output is 1 when the input is 0 and 0 when the input is 1.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	Input	Output	0	1	1	0				
Input	Output												
0	1												
1	0												
AND	The output is 1 only when both inputs are 1, otherwise the output is 0.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0 0</td><td>0</td></tr><tr><td>0 1</td><td>0</td></tr><tr><td>1 0</td><td>0</td></tr><tr><td>1 1</td><td>1</td></tr></table>	Input	Output	0 0	0	0 1	0	1 0	0	1 1	1
Input	Output												
0 0	0												
0 1	0												
1 0	0												
1 1	1												
OR	The output is 0 only when both inputs are 0, otherwise the output is 1.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0 0</td><td>0</td></tr><tr><td>0 1</td><td>1</td></tr><tr><td>1 0</td><td>1</td></tr><tr><td>1 1</td><td>1</td></tr></table>	Input	Output	0 0	0	0 1	1	1 0	1	1 1	1
Input	Output												
0 0	0												
0 1	1												
1 0	1												
1 1	1												
NAND	The output is 0 only when both inputs are 1, otherwise the output is 1.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0 0</td><td>1</td></tr><tr><td>0 1</td><td>1</td></tr><tr><td>1 0</td><td>1</td></tr><tr><td>1 1</td><td>0</td></tr></table>	Input	Output	0 0	1	0 1	1	1 0	1	1 1	0
Input	Output												
0 0	1												
0 1	1												
1 0	1												
1 1	0												
NOR	The output is 1 only when both inputs are 0, otherwise the output is 0.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0 0</td><td>1</td></tr><tr><td>0 1</td><td>0</td></tr><tr><td>1 0</td><td>0</td></tr><tr><td>1 1</td><td>0</td></tr></table>	Input	Output	0 0	1	0 1	0	1 0	0	1 1	0
Input	Output												
0 0	1												
0 1	0												
1 0	0												
1 1	0												
XOR	The output is 1 only when the two inputs have different value, otherwise the output is 0.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0 0</td><td>0</td></tr><tr><td>0 1</td><td>1</td></tr><tr><td>1 0</td><td>1</td></tr><tr><td>1 1</td><td>0</td></tr></table>	Input	Output	0 0	0	0 1	1	1 0	1	1 1	0
Input	Output												
0 0	0												
0 1	1												
1 0	1												
1 1	0												
XNOR	The output is 1 only when the two inputs have the same value, otherwise the output is 0.		<table><tr><th>Input</th><th>Output</th></tr><tr><td>0 0</td><td>1</td></tr><tr><td>0 1</td><td>0</td></tr><tr><td>1 0</td><td>0</td></tr><tr><td>1 1</td><td>1</td></tr></table>	Input	Output	0 0	1	0 1	0	1 0	0	1 1	1
Input	Output												
0 0	1												
0 1	0												
1 0	0												
1 1	1												

Introduction > Classical and Quantum Gates from Course 1

GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE	BLOCH SPHERE						
I Identity-gate: no rotation is performed.		$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$0\rangle$</td></tr><tr><td>$1\rangle$</td><td>$1\rangle$</td></tr></table>	Input	Output	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$	
Input	Output									
$ 0\rangle$	$ 0\rangle$									
$ 1\rangle$	$ 1\rangle$									
X gate: rotates the qubit state by π radians (180°) about the x-axis.		$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$1\rangle$</td></tr><tr><td>$1\rangle$</td><td>$0\rangle$</td></tr></table>	Input	Output	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$	
Input	Output									
$ 0\rangle$	$ 1\rangle$									
$ 1\rangle$	$ 0\rangle$									
Y gate: rotates the qubit state by π radians (180°) about the y-axis.		$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$i 1\rangle$</td></tr><tr><td>$1\rangle$</td><td>$-i 0\rangle$</td></tr></table>	Input	Output	$ 0\rangle$	$i 1\rangle$	$ 1\rangle$	$-i 0\rangle$	
Input	Output									
$ 0\rangle$	$i 1\rangle$									
$ 1\rangle$	$-i 0\rangle$									
Z gate: rotates the qubit state by π radians (180°) about the z-axis.		$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$0\rangle$</td></tr><tr><td>$1\rangle$</td><td>$- 1\rangle$</td></tr></table>	Input	Output	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$- 1\rangle$	
Input	Output									
$ 0\rangle$	$ 0\rangle$									
$ 1\rangle$	$- 1\rangle$									
S gate: rotates the qubit state by $\frac{\pi}{2}$ radians (90°) about the z-axis.		$S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$0\rangle$</td></tr><tr><td>$1\rangle$</td><td>$e^{i\frac{\pi}{2}} 1\rangle$</td></tr></table>	Input	Output	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$e^{i\frac{\pi}{2}} 1\rangle$	
Input	Output									
$ 0\rangle$	$ 0\rangle$									
$ 1\rangle$	$e^{i\frac{\pi}{2}} 1\rangle$									
T gate: rotates the qubit state by $\frac{\pi}{4}$ radians (45°) about the z-axis.		$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$0\rangle$</td></tr><tr><td>$1\rangle$</td><td>$e^{i\frac{\pi}{4}} 1\rangle$</td></tr></table>	Input	Output	$ 0\rangle$	$ 0\rangle$	$ 1\rangle$	$e^{i\frac{\pi}{4}} 1\rangle$	
Input	Output									
$ 0\rangle$	$ 0\rangle$									
$ 1\rangle$	$e^{i\frac{\pi}{4}} 1\rangle$									
H gate: rotates the qubit state by π radians (180°) about an axis diagonal in the x-z plane. This is equivalent to an X-gate followed by a $\frac{\pi}{2}$ rotation about the y-axis.		$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$0\rangle$</td><td>$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$</td></tr><tr><td>$1\rangle$</td><td>$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$</td></tr></table>	Input	Output	$ 0\rangle$	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$	$ 1\rangle$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$	
Input	Output									
$ 0\rangle$	$\frac{ 0\rangle + 1\rangle}{\sqrt{2}}$									
$ 1\rangle$	$\frac{ 0\rangle - 1\rangle}{\sqrt{2}}$									

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Controlled-NOT gate: apply an X-gate to the target qubit if the control qubit is in state $ 1\rangle$		$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$00\rangle$</td><td>$00\rangle$</td></tr><tr><td>$01\rangle$</td><td>$01\rangle$</td></tr><tr><td>$10\rangle$</td><td>$11\rangle$</td></tr><tr><td>$11\rangle$</td><td>$10\rangle$</td></tr></table>	Input	Output	$ 00\rangle$	$ 00\rangle$	$ 01\rangle$	$ 01\rangle$	$ 10\rangle$	$ 11\rangle$	$ 11\rangle$	$ 10\rangle$
Input	Output												
$ 00\rangle$	$ 00\rangle$												
$ 01\rangle$	$ 01\rangle$												
$ 10\rangle$	$ 11\rangle$												
$ 11\rangle$	$ 10\rangle$												
Controlled-phase gate: apply a Z-gate to the target qubit if the control qubit is in state $ 1\rangle$		$cZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	<table><tr><th>Input</th><th>Output</th></tr><tr><td>$00\rangle$</td><td>$00\rangle$</td></tr><tr><td>$01\rangle$</td><td>$01\rangle$</td></tr><tr><td>$10\rangle$</td><td>$10\rangle$</td></tr><tr><td>$11\rangle$</td><td>$- 11\rangle$</td></tr></table>	Input	Output	$ 00\rangle$	$ 00\rangle$	$ 01\rangle$	$ 01\rangle$	$ 10\rangle$	$ 10\rangle$	$ 11\rangle$	$- 11\rangle$
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RSA Cryptosystem, Factoring, and Shor's Algorithm > Quantum Fourier Transform



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